Dr. Michail Mamalakis

+44 7562576321 | mm2703@cam.ac.uk | LinkedIn | GitHub | Personal website | YouTube | ORCID: 0000-0002-4276-4119

Research Profile

- AI and computational-biology researcher developing explainable, robust and clinically relevant machine-learning methods for biomedical discovery and translational medicine.
- Research interests include foundation models, multimodal AI, medical-image analysis, single-cell and histology-informed modelling, protein language models, mechanistic interpretability and trustworthy AI.
- Current work focuses on paediatric brain cancer, neuroscience, multi-scale biomedical data integration and the use of explanations to identify clinically and biologically meaningful patterns.

Education

PhD in Computer Science — *Mar 2018 - Mar 2022*
University of Sheffield; major: deep learning and machine learning; minor: medical-image analysis — Sheffield, UK

MSc in Biomedical Engineering — *Aug 2015 - May 2017*
University of Patras — Achaia, Greece

MEng in Electrical Engineering — *Aug 2013 - May 2015*
University of Patras; minor: computer science — Achaia, Greece

BEng in Electrical Engineering — *Aug 2008 - May 2013*
University of Patras; minor: computer science — Achaia, Greece

Academic and Research Appointments

Assistant Research Professor — *Oct 2025 - Present*
University of Cambridge / CRUK Cambridge Institute — Cambridge, UK

- AI specialist working across neuroscience, paediatric and adult brain tumours, medical-image analysis and bioinformatics.
- Develops deep-learning, explainable-AI, machine-learning and software-engineering methods for biomedical research.

Affiliated Lecturer — *Oct 2025 - Present*
Department of Computer Science, University of Cambridge — Cambridge, UK

- Course director and lead lecturer for *Principles of AI-driven Neuroscience and Translational Biomedicine*: https://www.cl.cam.ac.uk/teaching/2526/L205/.
- Teaching covers deep learning, explainable AI, machine learning, neuroscience and bioinformatics.

Research Associate — *Jan 2024 - Sep 2025*
Department of Psychiatry, University of Cambridge / E-BRAIN WP2 Grant — Cambridge, UK

- AI specialist in neuroscience, brain-tumour research, medical-image analysis and computer vision.
- Developed deep-learning, explainable-AI, machine-learning and software-engineering methods for biomedical applications.

Research Associate — *Oct 2022 - Jan 2024*
Department of Psychiatry, University of Cambridge / MRC Grant — Cambridge, UK

- AI specialist in neuroscience, medical-image analysis and computer vision.
- Worked on deep learning, explainability, machine learning and software engineering for clinical and neuroscience datasets.

Research Associate — *Sep 2021 - Oct 2022*
IICD Department, University of Sheffield — Sheffield, UK

- AI specialist in medical-image analysis and computer vision.
- Worked on deep learning, explainability, uncertainty estimation, machine learning and software engineering.

Co-Founder and Researcher — *Dec 2016 - May 2018*
Patra/Achaia — Western Greece, Greece

- Developed medical signal-processing methods for wearable ECG monitoring, including noise reduction, ECG-monitoring architecture, accuracy evaluation and validation against clinical data.
- Worked as a medical signal-processing research engineer.

Projects and Research

scRNA-seq and Histology-Based Multi-Scale AI in Paediatric Brain Cancer *Sep 2025 - Present*
GNNs, LLMs, CNNs, evaluation protocols, representation learning, reasoning, explainable AI and mechanistic interpretability.
University of Cambridge, CRUK-CI.

MRI-Imaging Foundation Model for Neuroscience *Sep 2025 - Present*
Diffusion transformers, Swin-UNET, CNNs, evaluation protocols, explainable AI and mechanistic interpretability.
University of Cambridge.

Explainable Foundation Model for Neuroscience *Dec 2024 - Present*
GNNs, LLMs, CNNs, genetics, evaluation protocols, explainable AI and mechanistic interpretability.
University of Cambridge.

Protein and Clinical Language Models with Mechanistic Interpretability *Dec 2024 - Present*
Evaluation protocols, explainable AI, protein sequences and mechanistic interpretability.
University of Cambridge; Shanghai Jiao Tong University.

E-BRAIN WP2: Brain-Tumour Dataset Collection and Evaluation *Mar 2024 - Mar 2025*
Computer vision, federated learning, explainable AI and brain data science.
University of Cambridge.

Explainable AI Frameworks to Uncover New Patterns in Neuroscience *Oct 2022 - Present*
Explainable AI, uncertainty estimation, 3D CNNs, 3D vision transformers, GNNs and self-supervised learning.
University of Cambridge; related publications: [1], [2], [5], [6], [17].

Protein Language Model for Protein-Sequence Design *Oct 2023 - Dec 2024*
Evaluation protocols, explainable AI, protein sequences and scientific writing.
University of Cambridge; Johns Hopkins University; Shanghai Jiao Tong University; related publications: [12], [13].

Detecting Lung Disease in 3D Space Using Uncertainty Estimation *Oct 2021 - Oct 2022*
Automatic pipeline development, 3D patch extraction, explainable AI and uncertainty estimation.
University of Sheffield; related publications: [4], [9].

Cross-University Collaboration for AI-Driven Diagnostic Solutions for COVID-19 *Jun 2019 - Jun 2022*
Medical image-analysis research scientist and co-founder.
Universities of Sheffield, Brighton, Glasgow, Oxford and Lincoln; related publication: [9].

PhD Thesis in Computer Science / Computer Vision *Mar 2018 - Oct 2022*
Medical-image analysis and automatic segmentation of the left ventricle with scar using machine learning and deep learning.
University of Sheffield; related publications: [7], [10], [11].

MSc Thesis in Biomedical Engineering *Sep 2015 - Jun 2017*
Neuromuscular excitation modelling and simulation of the human knee using FEA dynamics and OpenSim.
University of Patras.

Undergraduate Thesis in Electrical Engineering and Computer Science *Sep 2013 - Jun 2015*
Electromagnetic dynamic study of cage-rotor AC machines and signal diagnostic errors using FEM.
University of Patras.

PUBLICATIONS

Asterisks denote equal contribution where indicated by the publication record. Original publication numbers are retained because they are cited in the Projects and Research section.

Selected Top Publications (Last Five Years)

S1 **Mamalakis M., Mamalakis A., et al.** *Solving the Enigma: Deriving Optimal Explanations of Deep Networks*. *AI Open*, 2025. https://doi.org/10.1016/j.aiopen.2025.02.001

S2 **Mamalakis M., Dwivedi K., Sharkey M., et al.** *A Transparent Artificial Intelligence Framework to Assess Lung Disease in Pulmonary Hypertension*. *Scientific Reports*, 13, 3812 (2023). https://doi.org/10.1038/s41598-023-30503-4

S3 **Mamalakis M., Garg P., Nelson T., et al.** *Artificial Intelligence Framework with Traditional Computer Vision and Deep Learning Approaches for Optimal Automatic Segmentation of Left Ventricle with Scar*. *Artificial Intelligence in Medicine*, 143, 102610 (2023). https://doi.org/10.1016/j.artmed.2023.102610

S4 **Abulikemu, Azevedo, Mamalakis M.*, Suckling J.*** *Geometry-Guided Generative Representation for Functional Brain Graphs*. Accepted at ICML 2026. https://icml.cc/virtual/2026/poster/62212

Selected for the University of Birmingham application.

Published peer-reviewed articles and conference papers

[2] Mamalakis M., Vareilles H., Al-Manea A., et al. *An Explainable Three-Dimensional Framework to Uncover Learning Patterns: A Unified Look in Variable Sulci Recognition*, Artificial Intelligence in Medicine, Elsevier, Preprint:https://doi.org/10.48550/arXiv.2309.00903.

[3] Mamalakis M., Macfarlane S. C., Notley S., et al. *Deep Multi-Attention Channels Network to Detect Metastasizing Cells in Fluorescence Microscopy Images*. *Computers in Biology and Medicine*, Elsevier. Preprint: https://doi.org/10.48550/arXiv.2309.00911.

[4] **SELECTED TOP PUBLICATION**
Mamalakis M., Dwivedi K., Sharkey M., et al. *A Transparent Artificial Intelligence Framework to Assess Lung Disease in Pulmonary Hypertension*. *Scientific Reports*, 13, 3812 (2023). https://doi.org/10.1038/s41598-023-30503-4.

[5] **SELECTED TOP PUBLICATION**
Mamalakis M., Mamalakis A., et al. *Solving the Enigma: Deriving Optimal Explanations of Deep Networks*. *AI Open*, Elsevier, 2025. https://doi.org/10.1016/j.aiopen.2025.02.001.

[7] **SELECTED TOP PUBLICATION**
Mamalakis M., Garg P., Nelson T., et al. *Artificial Intelligence Framework with Traditional Computer Vision and Deep Learning Approaches for Optimal Automatic Segmentation of Left Ventricle with Scar*. *Artificial Intelligence in Medicine*, 143, 102610, 2023. ISSN 0933-3657. https://doi.org/10.1016/j.artmed.2023.102610.

[8] Mamalakis M., Banerjee A., Roy S., et al. *Deep Multi-Metric Training: The Need for Multi-Metric Curve Evaluation to Avoid Weak Learning*. *Neural Computing and Applications*, 2024. https://doi.org/10.1007/s00521-024-10182-6.

[9] Mamalakis M., Swift A. J., Vorselaars B., et al. *DenResCov-19: A Deep Transfer Learning Network for Robust Automatic Classification of COVID-19, Pneumonia and Tuberculosis from X-Rays*. *Computerized Medical Imaging and Graphics*, 94, 102008, 2021. ISSN 0895-6111. https://doi.org/10.1016/j.compmedimag.2021.102008.

[10] Mamalakis M., Garg P., Nelson T., et al. *MA-SOCRATIS: An Automatic Pipeline for Robust Segmentation of the Left Ventricle and Scar*. *Computerized Medical Imaging and Graphics*, 93, 101982, 2021. ISSN 0895-6111. https://doi.org/10.1016/j.compmedimag.2021.101982.

[11] Mamalakis M., Garg P., Nelson T., et al. *Automatic Development of 3D Anatomical Models of Border Zone and Core Scar Regions in the Left Ventricle*. *Computerized Medical Imaging and Graphics*, 103, 102152, 2023. ISSN 0895-6111. https://doi.org/10.1016/j.compmedimag.2022.102152.

[12] Shen Y.*, Chen Z.*, Mamalakis M.*, et al. *A Finetuning Dataset and Benchmark for Large Language Models on Protein Sequence Understanding*. IEEE BIBM Conference 2024. http://arxiv.org/abs/2406.05540.

[13] Shen Y.*, Chen Z.*, Mamalakis M.*, et al. *TourSynbio: A Multi-Modal Large Model and Agent Framework to Bridge Text and Protein Sequences for Protein Engineering*. IEEE BIBM 2024. https://arxiv.org/abs/2408.15299.

[16] Sansone G., Poologaindran A., Romero-Garcia R., Mamalakis M., et al. *Longitudinal Modifications of Gradients and Graph-Theory Measures of Functional MRI Connectivity in Glioma Patients as Potential Markers of Post-Surgical Long-Term Cognitive Outcome*. *Neuro-Oncology*, 2024.

Accepted papers

[1] Mamalakis M., Vareilles H., et al. *The Explanation Necessity for Healthcare AI*. Accepted at IEEE CITREx Conference 2025. Preprint: https://arxiv.org/abs/2406.00216.

[18] **SELECTED TOP PUBLICATION**
Abulikemu, Azevedo, Mamalakis M.*, Suckling J.* *Geometry-Guided Generative Representation for Functional Brain Graphs*. Accepted at ICML 2026. https://icml.cc/virtual/2026/poster/62212.

Preprints

[6] Mamalakis M., et al. *Contrastive-Adversarial and Diffusion: Exploring Pre-Training and Fine-Tuning Strategies for Sulcal Identification*. arXiv, 2024. https://arxiv.org/abs/2405.19204.

[15] Jimenez-Mesa C., Wan Y., Sansone G., Martinez-Murcia F. J., Ramirez J., Liò P., Gorriz J. M., Price S. J., Suckling J., Mamalakis M. *Uncovering Neuroimaging Biomarkers of Brain Tumor Surgery with AI-Driven Methods*. arXiv, 2025. https://arxiv.org/abs/2507.04881.

[17] Mamalakis M., et al. *A Monosemantic Attribution Framework for Stable Interpretability in Clinical Neuroscience Large Language Models*. arXiv, 2026. https://www.arxiv.org/abs/2601.17952.

Manuscripts under review

[14] Cosentino C., Sun Z., Azevedo T., Abulikemu S., Bethlehem R., Marozzo F., Liò P., Mamalakis M. *Explainable Deep Learning for Alzheimer's Disease Diagnosis: An Integrated Clinical Data-Driven Approach*. Under review at *IEEE Transactions on Neural Networks and Learning Systems*.

Work in preparation

[21] Mamalakis M. et al. *A Continually Expandable Foundation Model for Brain MRI*. Work in preparation.

[22] Mamalakis M. et al. *Spatial Cell Embedding and Gene Semantic Fusion*. Work in preparation.

[23] Mamalakis M. et al. *Assessing the Importance of Gene Semantic Fusion in Murine Ependymoma*. Work in preparation.

Publication profile: https://scholar.google.com/citations?user=SbRBtk4AAAAJ&hl=en&oi=ao.

Scientific Reviewer and Examiner Experience

Internal Examiner, PhD Viva *Apr 2026*
Cambridge School of Clinical Medicine Cambridge, UK

Examiner, PhD First-Year Report *Dec 2025*
Clinical Neurosciences, University of Cambridge Cambridge, UK

External Examiner, MSc Viva *Mar 2026*
Department of Computer Science, EPFL Lausanne, Switzerland

Official Reviewer, 2nd Explainable AI World Conference *2024 - 2025*
https://xaiworldconference.com/2024/

Official Reviewer, MICCAI *2022 - 2026*
https://miccai.org/

Official Reviewer, IEEE Transactions on Medical Imaging *2023 - 2026*
IEEE TMI

Official Reviewer, IEEE Transactions on Image Processing *2026*
IEEE TIP

Official Reviewer, Elsevier Journals *2021 - 2026*
Medical imaging, computer vision and biomedical AI journals

EPSRC/UKRI Grant-Application Reviewer *2023 - 2026*
Reviewed four applications UK

EPSRC/UKRI Grant-Application Reviewer, UK AISI Systemic AI Safety Call *2024 - 2025*
Reviewed applications for systemic AI safety research UK

Seminars and Invited Presentations

CRUK-CI Lunch Seminar Series *Mar 2026*
Foundation Models Across Biological Scales: From Imaging to scRNA-seq and Back

Presented work on bioinformatics foundation models in imaging and omics, and on the use of explanations to identify new markers. Cambridge, UK.

Invited Speaker, IJCNN 2025: AI for a Cooler Planet *Jul 2025*
Explainable and Interpretable AI: From Interpretation to New Pattern Discovery

Presented work on explainable AI and the use of explanations to identify new markers. Rome, Italy.

Presenter, RadNet Cambridge and CRUK Children's Brain Tumour Centre of Excellence *Feb 2025*
Explainable and Interpretable AI: Enhancing Trust and Identifying Patterns in Healthcare and Neuroscience

Presented to early-career researchers across cancer research, with a focus on brain cancer and neuroscience. Cambridge, UK.

Presenter, Cambridge Accelerate Lunchtime Seminar Series *Feb 2024*
3D Explainability Frameworks

Presented work on a 3D explainability framework. Cambridge, UK.

Presenter, Cambridge Centre for AI in Medicine Summer School *Sep 2023*
Medical Imaging and 3D Explainability

Presented work on medical imaging and 3D explainability (summer school link). Cambridge, UK.

Organiser and Presenter, CHIA and AstraZeneca Workshop on Trustworthy AI in Imaging *Sep 2023*
Trustworthy AI in Imaging

Organised and presented at a medical-imaging AI workshop (workshop link). Cambridge, UK.

Lecturer, Sano/Insigneo Researcher Training Seminar *Oct 2022*
Medical-Imaging AI, Explainability and Uncertainty Estimation

Delivered a one-hour lecture on medical-imaging AI, explainability and uncertainty estimation. Sheffield, UK.

Teaching Experience

L205 ACS MPhil Course, Department of Computer Science *Lent 2026*
Course director, main lecturer, organiser and marker University of Cambridge, UK

Java Programming, Department of Computer Science *2 semesters*
Lab demonstrator and assessment marker University of Sheffield, UK

COM6103 Team Software Project, Department of Computer Science *2 semesters*
Agile methods, Scrum and team supervision University of Sheffield, UK

COM161/COM160 Introduction to Programming, Department of Computer Science *2 semesters*
Tutorial teaching University of Sheffield, UK

COM4509/6509 Machine Learning and AI, Department of Computer Science *1 semester*
Lab demonstrator and assessment marker University of Sheffield, UK

ACS341 Machine Learning, Department of Automatic Control and Systems Engineering — *2 semesters*
Lab demonstrator, teaching assistant and assessment marker — University of Sheffield, UK

ACS233 C++ Programming, Department of Automatic Control and Systems Engineering — *2 semesters*
Lab demonstrator and teaching assistant — University of Sheffield, UK

Funding and Grant Experience

EPSRC Access to High Performance Computing Facilities — *Apr 2024*
Awarded — UK

UKRI Access to High Performance Computing Facilities, APP94963 — *Dec 2025*
Awarded for *From MRI to Multi-Omics: An Explainable Foundation Model for Brain Cancer* — UK

BioBANK-DATASET access, Collaborator , ID759242 — *Mar 2026*
Awarded for *Personalized prediction of cognitive impairment among patients with cardiovascular disease using known risk factors, omics, and imaging- a multicohort analyses within the DORIAN GRAY project* — UK

BioBANK-DATASET access, Collaborator , ID20904 — *Mar 2026*
Awarded for *Genetic and phenotypic investigation of neural brain network connectivity* — UK

Honours and Awards

IEEE TMI Distinguished Reviewer, Gold Level — *Oct 2024*
Recognised after completing nine reviews between Sep 2023 and Oct 2024

Top-10 and Top-5 Team Placements, AIIB MICCAI Challenge 2023 — *Aug 2023*
Led a team of PhD students in two AI/ML medical computer-vision challenge tasks: airway-lung segmentation and survival prediction

CRUK Cambridge Centre Training Programme Postdoctoral Mini-Grant — *May 2023*
Shortlisted project for undergraduate selection: AI/ML medical imaging for brain tumours and federated learning (programme link)

Winning Team, AI vs COVID-19 UK Hackathon — *Apr 2020*
Member of the winning team among 20 participating teams in a high-profile multi-day virtual hackathon

University of Sheffield Department of Computer Science PhD Scholarship — *Apr 2018*
42-month PhD scholarship

Memberships and Affiliations

Insigneo Institute — *2018 - 2022*
Member as a PhD student and postdoctoral researcher at the University of Sheffield

Centre for Human-Inspired Artificial Intelligence — *2023 - Present*
Postdoctoral member, University of Cambridge

Cambridge Centre for AI in Medicine — *2024 - Present*
Postdoctoral member, University of Cambridge

Cambridge Neuroscience — *2022 - Present*
Member, University of Cambridge (member profile)

CRUK Children's Brain Tumour Centre of Excellence — *2025 - Present*
Postdoctoral Transition Fellow (centre website)

Editorial Experience

Guest Editor
Computerized Medical Imaging and Graphics, Elsevier
Special issue: *Computerized Approaches for the Development of AI-Based Clinical Decision-Support Systems Integrated into Clinical Practice* (special issue link).